

Program 9 : Organize k8s pods using labels and namespaces

prg9

----- namespace.yaml

----- pods.yaml

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ ls
namespaces.yaml  pods.yaml
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ cat namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev

apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ cat pods.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl apply -f namespaces.yaml
namespace/prod created
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl apply -f pods.yaml
pod/backend-pod created
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ █
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-h8wjc  1/1     Running   1 (4h8m ago)  4h40m
nginx-deployment-6f9664446b-xpszz  1/1     Running   1 (4h8m ago)  4h40m
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl get nodes
```

```
NAME      STATUS   ROLES    AGE     VERSION
minikube   Ready    control-plane  4h41m   v1.34.0
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl get deployments
```

```
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           4h40m
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl get svc
```

```
NAME           TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes     ClusterIP   10.96.0.1    <none>        443/TCP    4h42m
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl get services
```

```
NAME           TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes     ClusterIP   10.96.0.1    <none>        443/TCP    4h42m
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl get pods -n prod
```

```
NAME      READY   STATUS    RESTARTS   AGE
backend-pod  1/1     Running   0           4m
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ kubectl get pods -n prod -l app=backend
```

```
NAME      READY   STATUS    RESTARTS   AGE
backend-pod  1/1     Running   0           4m4s
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg9$ █
```

```
nginx-deployment-7c79c4bf97-jrqlh  1/1     Running   1 (89m ago)  95m
nginx-deployment-7c79c4bf97-p9rzc  0/1     ImagePullBackOff  0           95m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-7c79c4bf97-jrqlh  1/1     Running   1 (90m ago)  96m
nginx-deployment-7c79c4bf97-p9rzc  0/1     ImagePullBackOff  0           96m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-7c79c4bf97-jrqlh  1/1     Running   1 (90m ago)  97m
nginx-deployment-7c79c4bf97-p9rzc  1/1     Running   1 (90m ago)  97m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get nodes
NAME      STATUS   ROLES    AGE     VERSION
minikube   Ready    control-plane  105m   v1.28.3
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get svc
NAME           TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes     ClusterIP   10.96.0.1    <none>        443/TCP    105m
nginx-service  NodePort    10.96.147.46 <none>        80:32410/TCP  97m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods -n dev
NAME      READY   STATUS    RESTARTS   AGE
backend-pod  1/1     Running   2 (91m ago)  103m
frontend-pod  1/1     Running   2 (91m ago)  103m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods -n prod
NAME      READY   STATUS    RESTARTS   AGE
backend-pod  1/1     Running   2 (91m ago)  103m
frontend-pod  1/1     Running   2 (91m ago)  103m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods -n prod -l app=frontend
NAME      READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   2 (92m ago)  104m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods -n prod -l app=backend
NAME      READY   STATUS    RESTARTS   AGE
backend-pod  1/1     Running   2 (92m ago)  104m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods -n dev -l app=backend
NAME      READY   STATUS    RESTARTS   AGE
backend-pod  1/1     Running   2 (92m ago)  104m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ kubectl get pods -n dev -l app=frontend
NAME      READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   2 (92m ago)  104m
1RV24MC093_SANDARSH_J_N@ubuntu:~/devops$ █
```